

Symmetric and nested subcases of the Gaussian RSSI question

Run fa_banach_001, arXiv:2602.07981, candidate partial result

Source and Question

The source paper is E. Milman, S. Nakamura, and H. Tsuji, *The Gaussian Conjugate Rogers–Shephard Inequality*, arXiv:2602.07981 [1]. After proving the Gaussian conjugate Rogers–Shephard inequality and the extension

$$\gamma(K)\gamma(L) \leq \gamma(K \cap L)\gamma(aK + bL)$$

for $|a|, |b| \geq 1$ and $|a + b| \geq 1$, the authors ask whether the case $a = 1, b = -1$ also holds:

$$\gamma(K)\gamma(L) \leq \gamma(K \cap L)\gamma(K - L) \quad (\text{GRSI})$$

for all convex $K, L \subset \mathbb{R}^n$ with non-empty interior and Gaussian barycenters at the origin.

Theorem 1.3. *Let $K, L \subset \mathbb{R}^n$ denote convex sets with non-empty interior and Gaussian barycenters at the origin. Then for all $|a|, |b| \geq 1$ so that $|a + b| \geq 1$,*

$$\gamma(K)\gamma(L) \leq \gamma(K \cap L)\gamma(aK + bL), \quad (1.10)$$

with equality when $|a + b| > 1$ if and only if (1.7) holds.

Since K, L are inclusion-wise monotone under scaling, and since the Gaussian measure is invariant under reflection about the origin, the only two interesting cases above (from which everything else follows) are:

- (1) $a = b = 1$, yielding the GCRSI (1.8).
- (2) $a = 2$ and $b = -1$, which in general (for non-origin-symmetric L) is incomparable to (1.8); if $a > 2$, we also get the equality case.

It would be very interesting to obtain the case when $a = 1$ and $b = -1$, since this would recover in the scaling limit the RSSI (1.2) instead of its conjugate version (1.9). Unfortunately, our proof does not work in this case, because we reduce the problem to a certain Gaussian saturation sufficient condition, which turns out to simply be false. Consequently, we leave this case as an interesting open problem: under the assumptions of Theorem 1.3 does it hold that

$$\gamma(K)\gamma(L) \leq \gamma(K \cap L)\gamma(K - L) ? \quad (1.11)$$

We can also obtain the following conjugate version of (1.1).

Theorem 1.4 (Conjugate Milman–Pajor Inequality (CMPI)). *Let $K, L \subset \mathbb{R}^n$ denote convex sets with non-empty interior and Gaussian barycenters at the origin. Then for all $a, b \in \mathbb{R}$*

Result

Status. This packet proves narrow subcases of the source’s open problem, not the full Gaussian $K - L$ inequality.

Theorem 1. *Let $K, L \subset \mathbb{R}^n$ be convex sets with non-empty interior and Gaussian barycenters at the origin. Then the Gaussian $K - L$ inequality*

$$\gamma(K)\gamma(L) \leq \gamma(K \cap L)\gamma(K - L)$$

holds in each of the following cases:

1. *one of K, L is centrally symmetric;*
2. *$K \subset L$ or $L \subset K$.*

Proof Intuition

The centrally symmetric case is a sign-flip consequence of the source paper's already-proved $K + L$ theorem. If, for example, $K = -K$, apply the source theorem to K and $-L$. Reflection preserves Gaussian measure, and the symmetry of K converts the intersection $K \cap (-L)$ back to $K \cap L$ in measure.

The nested case is even simpler. Gaussian barycenter zero forces the origin to lie in each closed convex set. If $K \subset L$, then $K \cap L = K$ and $-L \subset K - L$, so $\gamma(K - L) \geq \gamma(L)$. The case $L \subset K$ is symmetric.

Lemma 1. *Let $K \subset \mathbb{R}^n$ be a convex set with non-empty interior and finite Gaussian measure. If the Gaussian barycenter of K is the origin, then $0 \in \overline{K}$. In particular, replacing K by $\overline{K}$ does not change its Gaussian measure or barycenter, and then $0 \in K$.*

Proof. If $0 \notin \overline{K}$, the strong separation theorem gives a vector $u \in \mathbb{R}^n$ and $c > 0$ such that $\langle u, x \rangle \geq c$ for all $x \in K$, after changing the sign of u if necessary. Integrating over K gives

$$\left\langle u, \int_K x d\gamma(x) \right\rangle = \int_K \langle u, x \rangle d\gamma(x) \geq c\gamma(K) > 0,$$

contradicting the assumption that the Gaussian barycenter is the origin. □

Proof of Theorem 1. First assume that $L = -L$. Then $K - L = K + L$, and the desired inequality is exactly the Gaussian conjugate Rogers–Shephard inequality proved in the source paper. If instead $K = -K$, apply the same source theorem to the pair $(K, -L)$, whose Gaussian barycenters are still at the origin:

$$\gamma(K)\gamma(-L) \leq \gamma(K \cap (-L))\gamma(K + (-L)).$$

By reflection invariance, $\gamma(-L) = \gamma(L)$, and since $K = -K$,

$$\gamma(K \cap (-L)) = \gamma((-K) \cap L) = \gamma(K \cap L).$$

Also $K + (-L) = K - L$. This proves the centrally symmetric case.

For the nested case, replace K, L by their closures; convex boundaries have Gaussian measure zero, and Lemma 1 gives $0 \in K \cap L$. If $K \subset L$, then $K \cap L = K$, and because $0 \in K$, $-L \subset K - L$. Hence

$$\gamma(K \cap L)\gamma(K - L) = \gamma(K)\gamma(K - L) \geq \gamma(K)\gamma(-L) = \gamma(K)\gamma(L).$$

If $L \subset K$, then $K \cap L = L$, and because $0 \in L$, $K \subset K - L$. Thus

$$\gamma(K \cap L)\gamma(K - L) = \gamma(L)\gamma(K - L) \geq \gamma(L)\gamma(K).$$

□

Corollary 1 (Parameter sign-flip with one symmetric set). *Let $a, b > 0$, and let $\alpha = \max(1/b, 1)$, $\beta = \max(1/a, 1)$. Assume that the source paper's $\sigma = +1$ inequality*

$$\gamma(K)\gamma(L) \leq \gamma(\alpha K \cap \beta L)\gamma(aK + bL)$$

holds for all centered convex K, L . Then the corresponding $\sigma = -1$ inequality

$$\gamma(K)\gamma(L) \leq \gamma(\alpha K \cap \beta L)\gamma(aK - bL)$$

holds whenever one of K, L is centrally symmetric.

Proof. If $L = -L$, then $aK - bL = aK + bL$. If $K = -K$, apply the $\sigma = +1$ inequality to K and $-L$, and use reflection invariance exactly as in the proof of Theorem 1. $\square$

Limitations

The argument gives no information about the genuinely nonsymmetric, nonnested case of the source problem. It also does not repair the Gaussian saturation method that the source paper explains fails for the $K - L$ case; it only bypasses the sign issue in situations where reflection converts $-L$ back to L in the relevant measure factors.

Verification and Novelty Check

The cheap run indexes were searched for arXiv:2602.07981, Gaussian conjugate Rogers–Shephard, the $K - L$ open problem, and related Rogers–Shephard keywords. No exact prior packet for these subcases was found. The centrally symmetric subcase is a direct corollary of the source paper's main theorem; the nested subcase is elementary.

Human Review Recommendation

Review as a valid but narrow partial result. The main checks are the reflection step in the one-symmetric-set case and the closure/origin argument in the nested case. Do not treat this as evidence for the full nonsymmetric Gaussian $K - L$ problem.

References

- [1] E. Milman, S. Nakamura, and H. Tsuji, *The Gaussian Conjugate Rogers–Shephard Inequality*, arXiv:2602.07981, 2026.